KAIZEN MATHS

Lesson starter

Basic Integration and Definite Integrals

Definite Integrals

Name _____

S1

Evaluate $\int_0^4 (4x^3 + 3x^2 - 3)\,dx.$

S2

Evaluate $\int_3^7 (3x^{\frac{5}{2}} + 4x^{\frac{5}{2}})\,dx.$

S3

Evaluate $\int_{-1}^1 (-2x^4 - 3x^2)\,dx.$

S4

Evaluate $\int_1^4 (-4x^{\frac{5}{2}} + 6x^{\frac{-5}{3}})\,dx.$

Show your working

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Show your working

KAIZEN MATHS

Exit ticket

Basic Integration and Definite Integrals

Definite Integrals

Name _____

E1

Evaluate $\int_{-1}^{2} (-5x^4 + 4x + 5)\,dx.$

E2

Evaluate $\int_{2}^{4} (3x^{\frac{4}{3}} + 6x^{\frac{-5}{3}})\,dx.$

E3

Evaluate $\int_{1}^{3} (4x^4 + 4x^2 - 4x)\,dx.$

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E1

Evaluate $\int_{-1}^{2} (-5x^4 + 4x + 5)\,dx.$

E2

Evaluate $\int_{2}^{4} (3x^{\frac{4}{3}} + 6x^{\frac{-5}{3}})\,dx.$

E3

Evaluate $\int_{1}^{3} (4x^4 + 4x^2 - 4x)\,dx.$

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Show your working

KAIZEN MATHS

Exit ticket

Basic Integration and Definite Integrals

Definite Integrals

Name _____

E1

Evaluate $\int_{-1}^{2} (-5x^4 + 4x + 5)\,dx.$

E2

Evaluate $\int_{2}^{4} (3x^{\frac{4}{3}} + 6x^{\frac{-5}{3}})\,dx.$

E3

Evaluate $\int_{1}^{3} (4x^4 + 4x^2 - 4x)\,dx.$

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Show your working

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Exit ticket

Basic Integration and Definite Integrals

Definite Integrals

Name _____

E1

Evaluate $\int_{-1}^{2} (-5x^4 + 4x + 5)\,dx.$

E2

Evaluate $\int_{2}^{4} (3x^{\frac{4}{3}} + 6x^{\frac{-5}{3}})\,dx.$

E3

Evaluate $\int_{1}^{3} (4x^4 + 4x^2 - 4x)\,dx.$

ALG-06EF-L2 · E

Show your working

Teacher answers & working

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S1

Evaluate $\int_0^4 (4x^3 + 3x^2 - 3)\,dx$.

Answer:

$= 308$

Working:

Step 1: First find an antiderivative F(x), then evaluate F(upper limit) - F(lower limit).

Use the reverse power rule. Add 1 to the power, then divide by the new power.

For the term $4x^3$, add 1 to the power.

Substitute into the reverse power rule.

Simplify the coefficient: $\frac{4}{4}x^4$

Write the simplified term.

For the term $3x^2$, add 1 to the power.

Substitute into the reverse power rule.

Simplify the coefficient: $\frac{3}{3}x^3$

Write the simplified term.

For the term -3, add 1 to the power.

Substitute into the reverse power rule.

Write the simplified term.

Combine the integrated terms.

Upper limit: substitute before calculating.

Evaluate the upper limit.

Lower limit: substitute before calculating.

Evaluate the lower limit.

Subtract lower from upper; round at the end.

Final answer: $= 308$

Teacher answers & working

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S2

Evaluate $\int_{3}^{7} (3x^{\frac{5}{2}} + 4x^{\frac{5}{2}})\,dx$.

Answer:

≈ 1721.45

Working:

Step 1: Integrate the fractional or negative powers first, then substitute the bounds.

Use the reverse power rule. Add 1 to the power, then divide by the new power.

Add 1 to the power.

Divide by the new power.

Simplify this term.

Add 1 to the power.

Divide by the new power.

Simplify this term.

Combine the integrated terms.

Upper limit: substitute before calculating.

Evaluate the upper limit.

Lower limit: substitute before calculating.

Evaluate the lower limit.

Subtract lower from upper; round at the end.

Final answer: ≈ 1721.45

Teacher answers & working

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S3

Evaluate $\int_{-1}^{1} (-2x^4 - 3x^2)\,dx$.

Answer:

≈ -2.80

Working:

Step 1: First find an antiderivative F(x), then evaluate F(upper limit) - F(lower limit).

Use the reverse power rule. Add 1 to the power, then divide by the new power.

For the term $-2x^4$, add 1 to the power.

Substitute into the reverse power rule.

Write the simplified term.

For the term $-3x^2$, add 1 to the power.

Substitute into the reverse power rule.

Simplify the coefficient: $\frac{-3}{3}x^3$

Write the simplified term.

Combine the integrated terms.

Upper limit: substitute before calculating.

Evaluate the upper limit.

Lower limit: substitute before calculating.

Evaluate the lower limit.

Subtract lower from upper; round at the end.

Final answer: ≈ -2.80

Teacher answers & working

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S4

Evaluate $\int_{1}^{4} (-4x^{\frac{5}{2}} + 6x^{\frac{-5}{3}})\,dx$.

Answer:

≈ -139.71

Working:

Step 1: Integrate the fractional or negative powers first, then substitute the bounds.

Use the reverse power rule. Add 1 to the power, then divide by the new power.

Add 1 to the power.

Divide by the new power.

Simplify this term.

Add 1 to the power.

Divide by the new power.

Simplify this term.

Combine the integrated terms.

Upper limit: substitute before calculating.

Evaluate the upper limit.

Lower limit: substitute before calculating.

Evaluate the lower limit.

Subtract lower from upper; round at the end.

Final answer: ≈ -139.71

Teacher answers & working

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E1

Evaluate $\int_{-1}^{2} (-5x^4 + 4x + 5)\,dx$.

Answer:

$= -12$

Working:

Step 1: First find an antiderivative F(x), then evaluate F(upper limit) - F(lower limit).

Use the reverse power rule. Add 1 to the power, then divide by the new power.

For the term $-5x^4$, add 1 to the power.

Substitute into the reverse power rule.

Simplify the coefficient: $\frac{-5}{5}x^5$

Write the simplified term.

For the term $4x$, add 1 to the power.

Substitute into the reverse power rule.

Simplify the coefficient: $\frac{4}{2}x^2$

Write the simplified term.

For the term 5, add 1 to the power.

Substitute into the reverse power rule.

Write the simplified term.

Combine the integrated terms.

Upper limit: substitute before calculating.

Evaluate the upper limit.

Lower limit: substitute before calculating.

Evaluate the lower limit.

Subtract lower from upper; round at the end.

Final answer: $= -12$

Teacher answers & working

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E2

Evaluate $\int_{2}^{4} (3x^{\frac{4}{3}} + 6x^{\frac{-5}{3}})\,dx$.

Answer:

≈ 28.27

Working:

Step 1: Integrate the fractional or negative powers first, then substitute the bounds.

Use the reverse power rule. Add 1 to the power, then divide by the new power.

Add 1 to the power.

Divide by the new power.

Simplify this term.

Add 1 to the power.

Divide by the new power.

Simplify this term.

Combine the integrated terms.

Upper limit: substitute before calculating.

Evaluate the upper limit.

Lower limit: substitute before calculating.

Evaluate the lower limit.

Subtract lower from upper; round at the end.

Final answer: ≈ 28.27

Teacher answers & working

Basic Integration and Definite Integrals · Definite Integrals

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E3

Evaluate $\int_{1}^{3} (4x^4 + 4x^2 - 4x)\,dx$.

Answer:

≈ 212.27

Working:

Step 1: First find an antiderivative F(x), then evaluate F(upper limit) - F(lower limit).

Use the reverse power rule. Add 1 to the power, then divide by the new power.

For the term $4x^4$, add 1 to the power.

Substitute into the reverse power rule.

Write the simplified term.

For the term $4x^2$, add 1 to the power.

Substitute into the reverse power rule.

Write the simplified term.

For the term $-4x$, add 1 to the power.

Substitute into the reverse power rule.

Simplify the coefficient: $\frac{-4}{2}$x^2

Write the simplified term.

Combine the integrated terms.

Upper limit: substitute before calculating.

Evaluate the upper limit.

Lower limit: substitute before calculating.

Evaluate the lower limit.

Subtract lower from upper; round at the end.

Final answer: ≈ 212.27